Day 1 – Session 2

2-Bit Mixed-Signal Full Adder

Integration of Digital and Analog Blocks for Arithmetic Operation

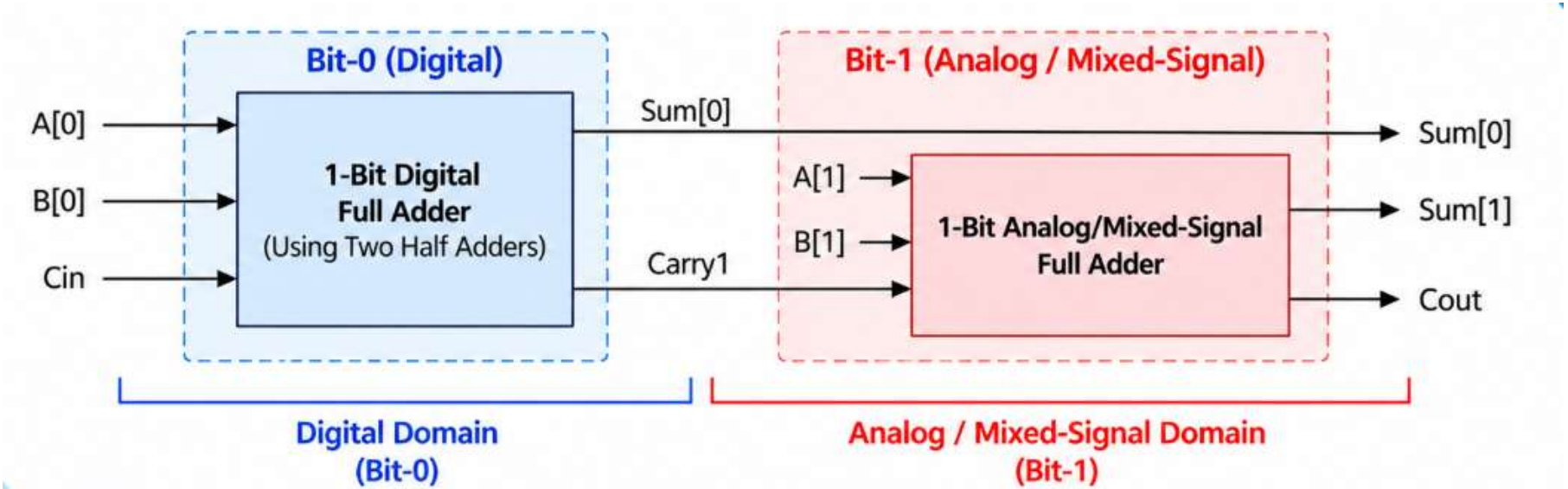
By:

SOUJANYA S

Ph.D Scholar

Cambridge Institute of Technology

Proposed 2-Bit Mixed-Signal Full Adder Architecture

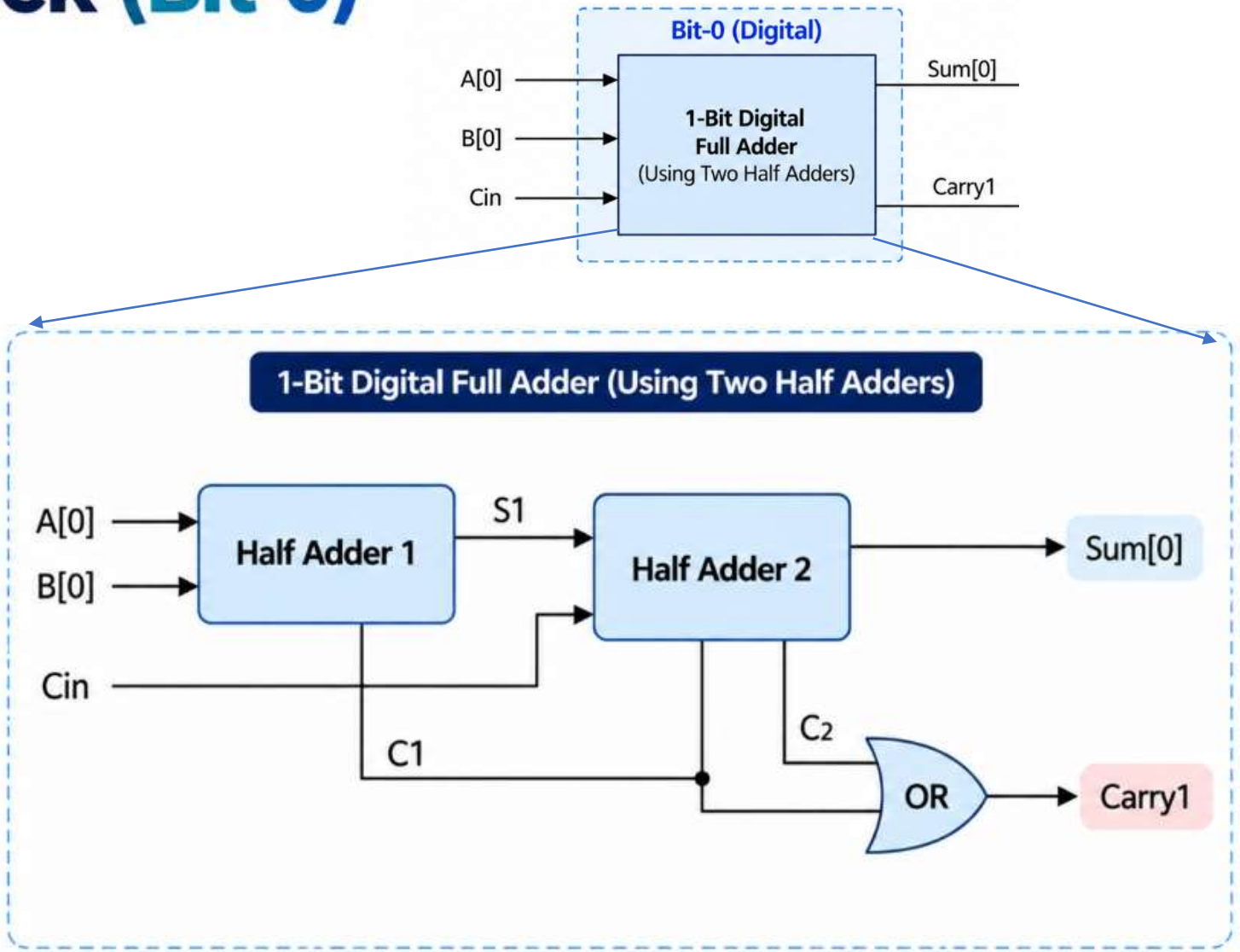


- Bit-0 is implemented using a digital full adder (constructed with two half adders).
- Sum[0] is generated at the output of the digital block.
- Only the carry (Carry1) from the digital block is passed to the analog/mixed-signal block.

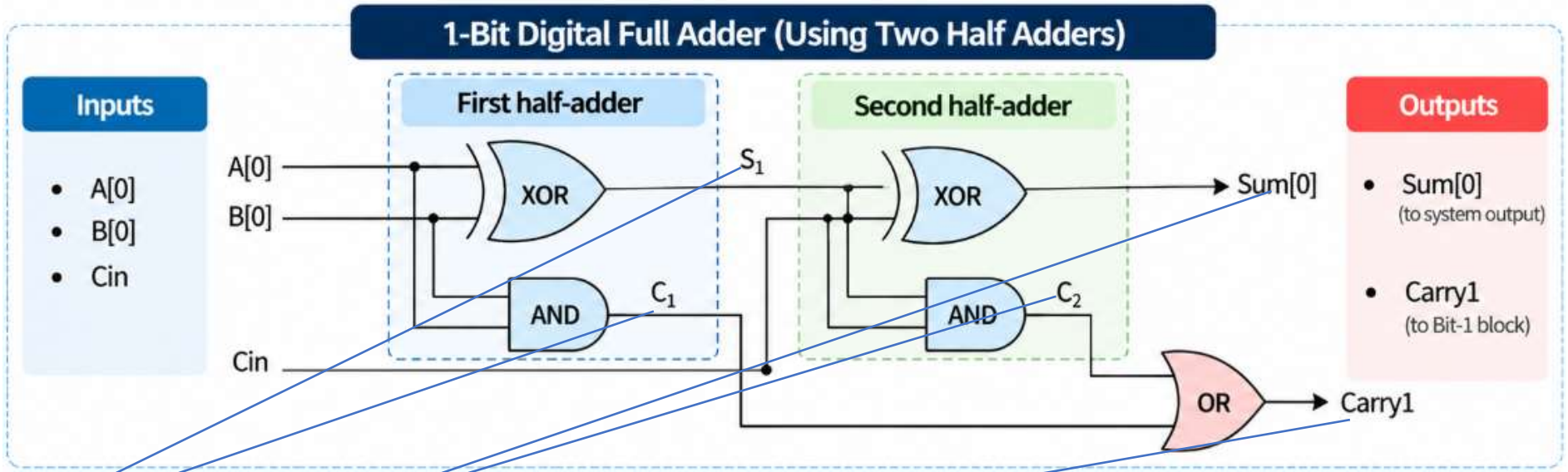
- Bit-1 is implemented using an analog/mixed-signal full adder.
- Outputs of the 2-bit adder: Sum[0], Sum[1] and Cout.

Digital Block (Bit-0)

- **Inputs**
  - A[0]
  - B[0]
  - Cin
- **Implementation**
  - 1-bit full adder using two half adders
- **Outputs**
  - Sum[0]
  - Carry1 (to Bit-1 block)



Digital Block (Bit-0) — Boolean Equations



Half Adder 1

$$S_1 = A_0 \oplus B_0$$

$$C_1 = A_0 \cdot B_0$$

Half Adder 2

$$Sum_0 = S_1 \oplus C_{in}$$

$$C_2 = S_1 \cdot C_{in}$$

Final Carry

$$Carry_1 = C_1 + C_2$$

Simplified Expressions

Sum Output

$$Sum_0 = A_0 \oplus B_0 \oplus C_{in}$$

Carry Output

$$Carry_1 = A_0B_0 + C_{in}(A_0 \oplus B_0)$$

ASIC Design Flow – RTL to GDSII

Front-End Design (Logic Design)

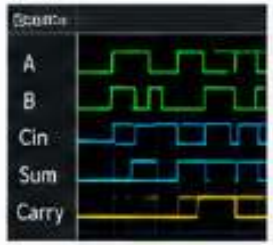
1. RTL Design

- Design Entry (Verilog/VHDL)
- Mixed-signal architecture (e.g., Full Adder)
- IP integration



2. Functional Verification

- Destbench development
- Simulation (Xcelium)
- Verify functionality



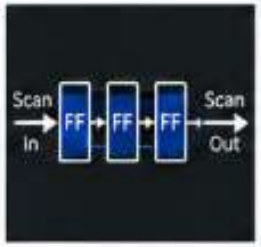
3. Logic Synthesis

- RTL to Gate-level
- Timing, Area, Power optimization
- Standard cell mapping



4. DFT (Scan Insertion)

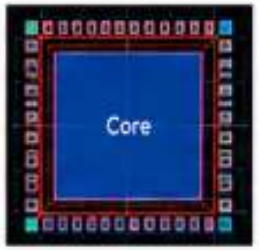
- Scan chain insertion
- Testability checks (ATPG)
- Geındard test netlists



Back-End Physical Design (Implementation)

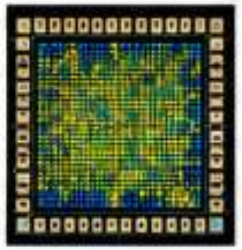
5. Floorplanning

- Define die size
- I/O pad placement
- Power planning
- Generate placement



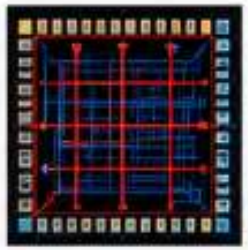
6. Placement

- Place standard cells
- Optimize timing, congestion



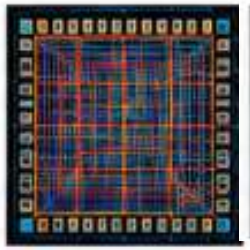
7. Clock Tree Synthesis (CTS)

- Build clock tree
- Optimize skew and latency



8. Routing

- Global & Detailed routing
- DRC clean routing



Verification & Tapeout

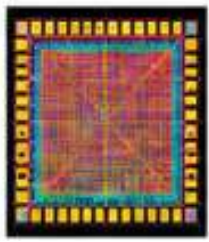
9. Physical Verification

- Design Rule Check (DRC)
- Layout vs Schematic (LVS)
- Parasitic Extraction (PEX)



10. GDSII / Tapeout

- Generate GDSII
- Signoff
- Submit for fabrication



SIMULATION

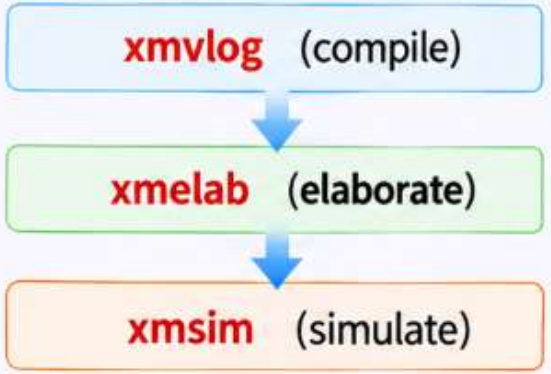
Command:

```
xrun -v top.v top_tb.v -access +rwc -gui
```

- xrun is the **single-step compile + elaborate + simulate** command in Cadence Xcelium.

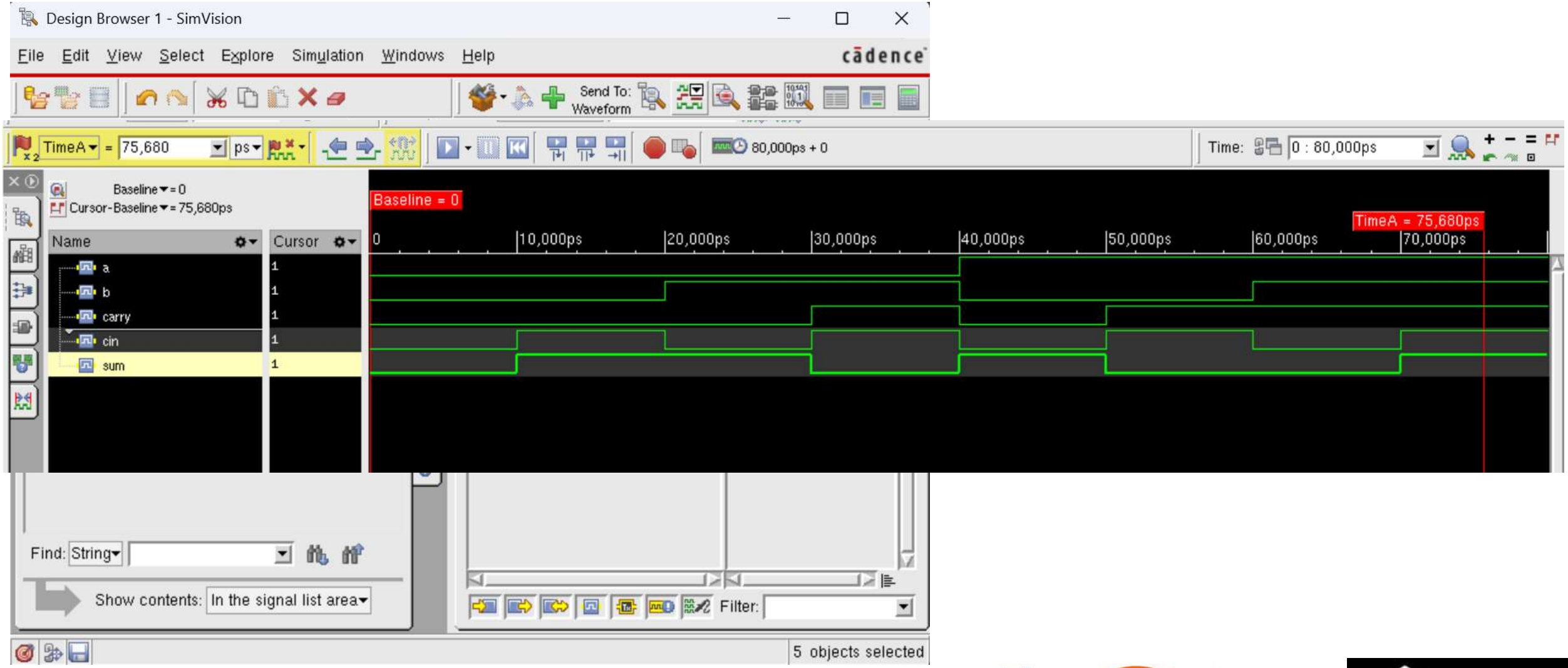
Letter	Meaning
r	**Read access** (Allows reading internal signals)
w	**Write access** (Allows writing to internal signals)
c	**Connectivity access** (Allows hierarchical access to internal design elements)

Instead of running separately:



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SIMULATION RESULTS

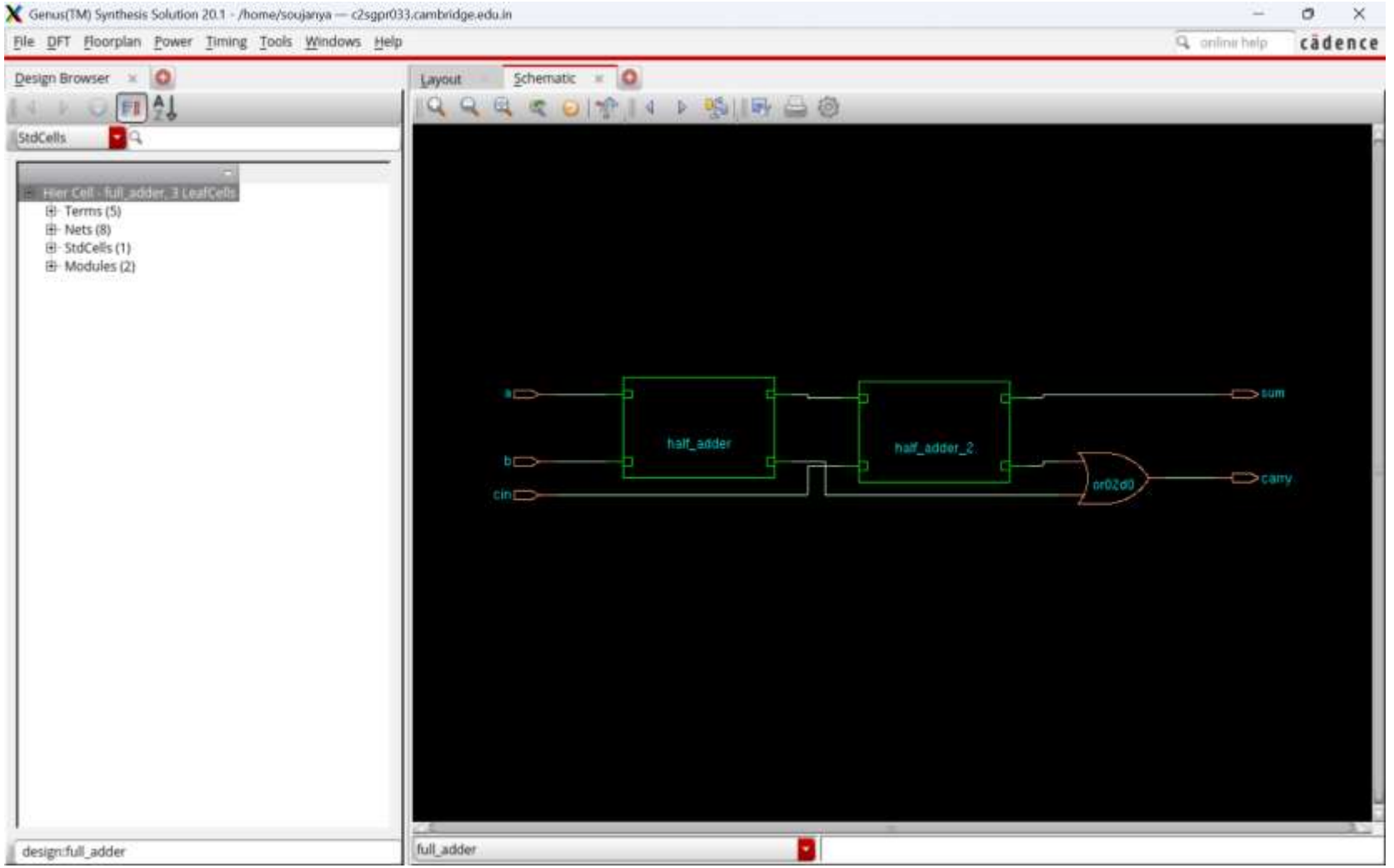


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SYNTHESIS

Genus tool used for synthesis

Source run.tcl



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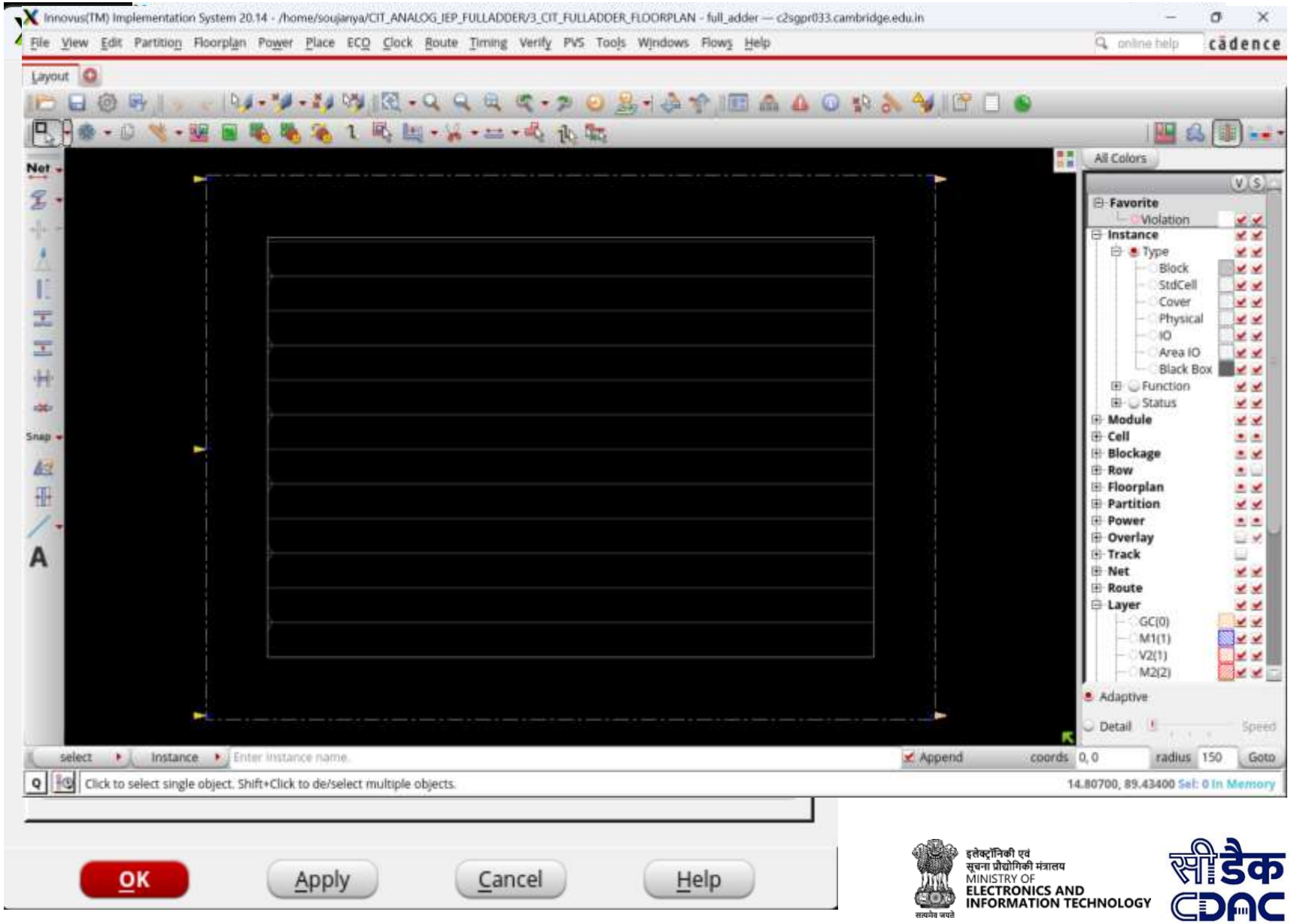
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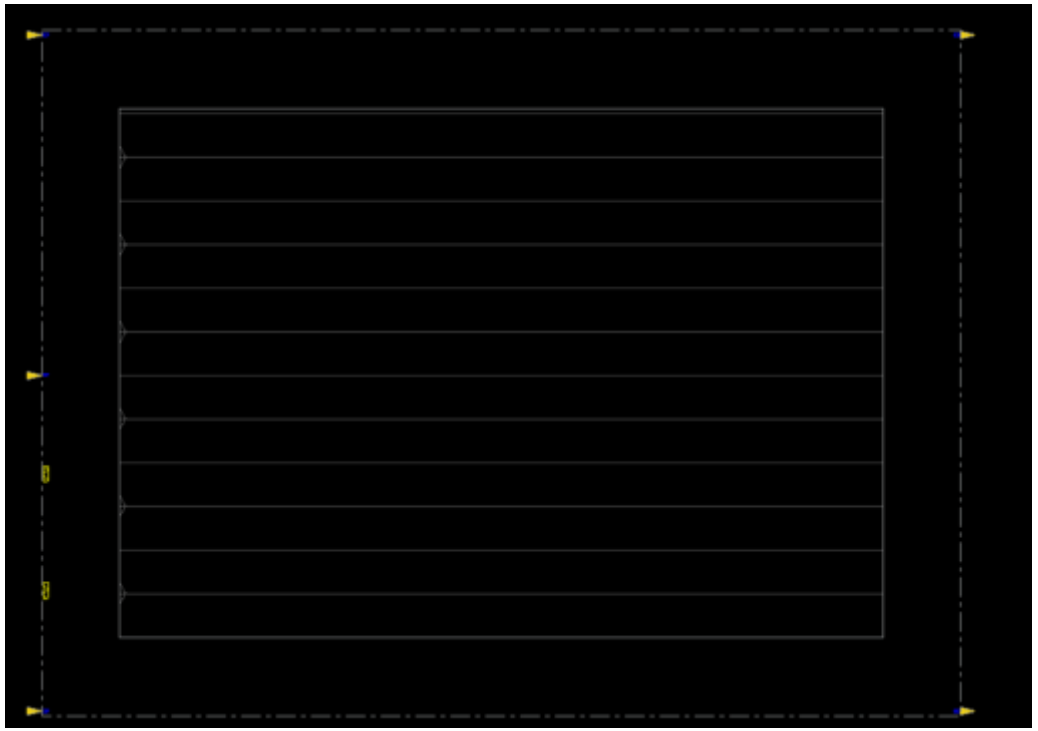
FLOORPLAN


Innovus(TM) Implementation System 20.14 - /home/soujanya/CIT_ANALOG_IEP_FULLADDER/3_CIT_FULLADDER_FLOORPLAN - full_adder — c2sgpr033.cambridge.edu.in
File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help
cadence
Layout
Net
Snap
All Colors
Favorite
Violation
Instance
Type
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Net
Route
Layer
GC(0)
M1(1)
V2(1)
M2(2)
Adaptive
Detail
Speed
select
Instance
Enter instance name.
Append
coords 0, 0
radius 150
Goto
Click to select single object. Shift+Click to de/select multiple objects.
14.80700, 89.43400 Sel: 0 In Memory
OK
Apply
Cancel
Help

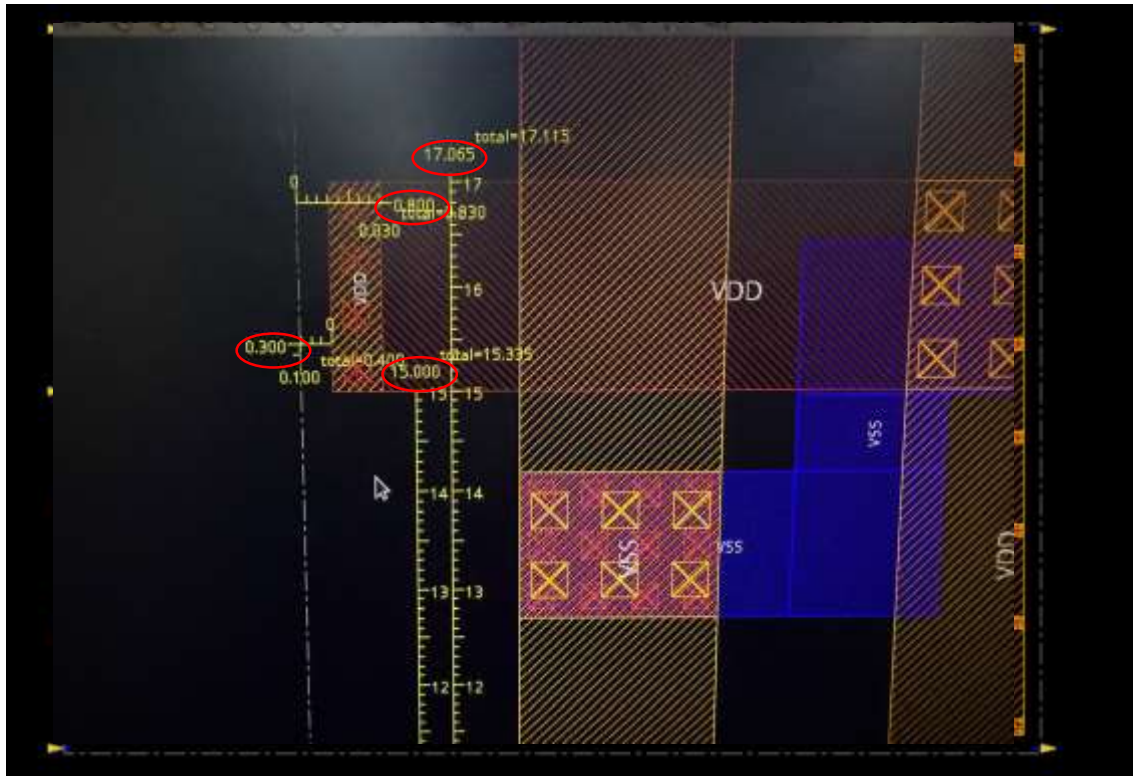
POWER PLAN

Source global net power connects in the Innovus

```
nts.tcl
innovus 1>
innovus 1> globalNetConnect VDD -type pgpin -pin VDD -all
innovus 2> globalNetConnect VSS -type pgpin -pin VSS -all
innovus 3> globalNetConnect VDD -type tiehi
innovus 4> globalNetConnect VSS -type tielo
innovus 5> createPGPin VDD -net VDD -geom M4 0.30 15.0 0.80 17.0
innovus 6> createPGPin VSS -net VSS -geom M4 0.30 30.0 0.80 32.0
innovus 7>
```

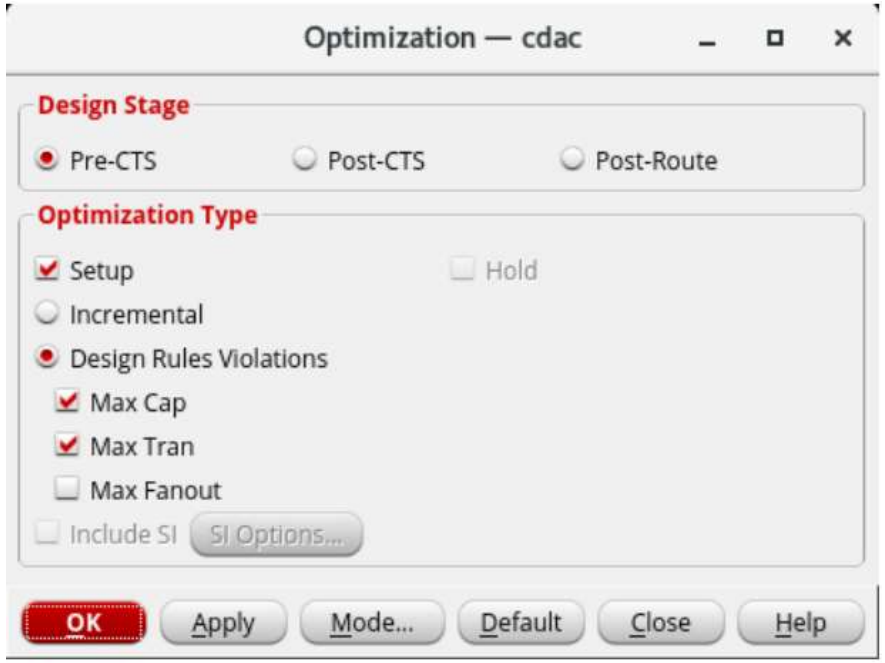


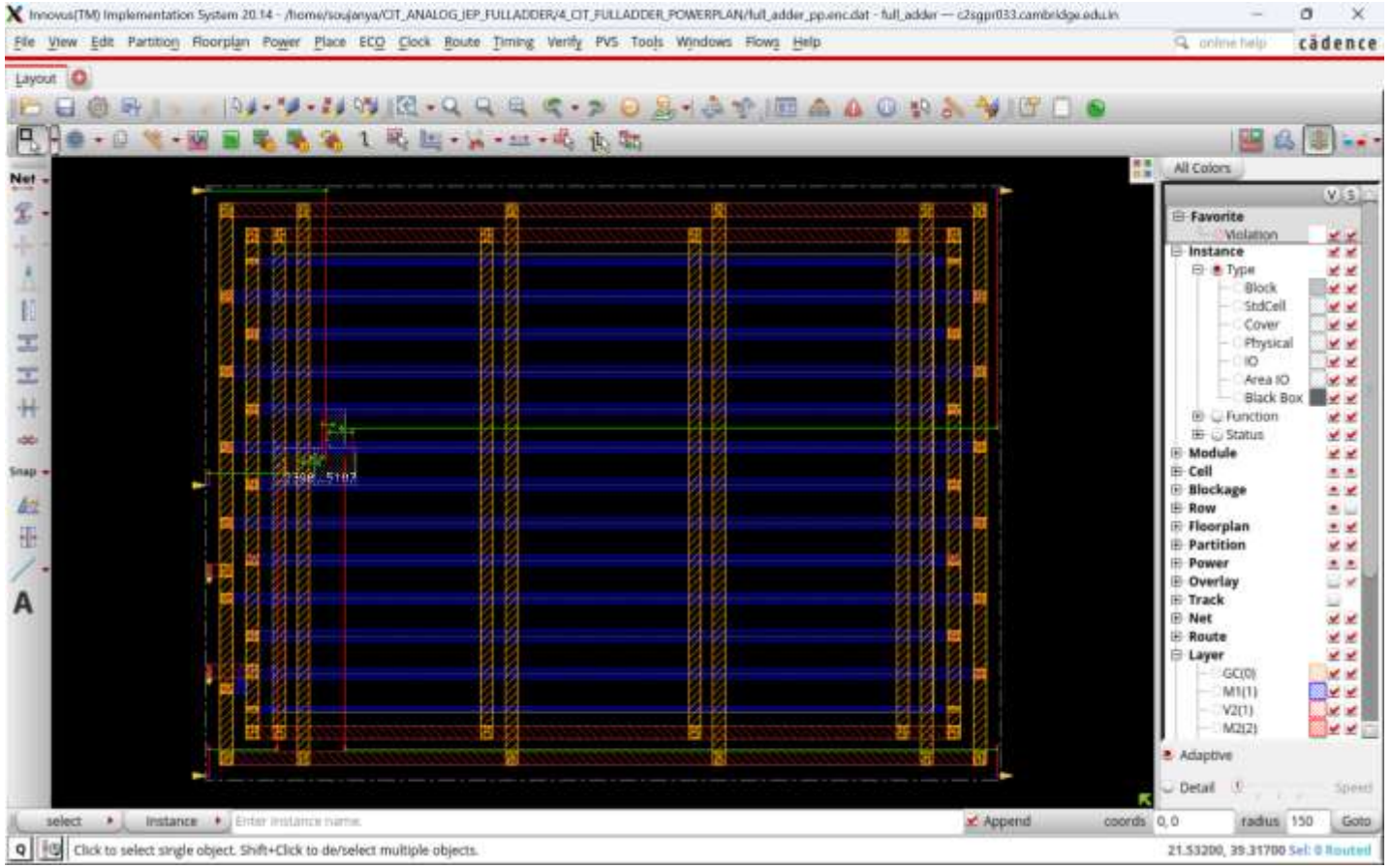
Power -> Power Planning -> Add Ring
Power -> Power Planning -> Add Stripes
Route → Special Route



PLACEMENT

- Place -> Place Standard Cell
- Timing -> Report Timing
- ECO -> Optimize Design



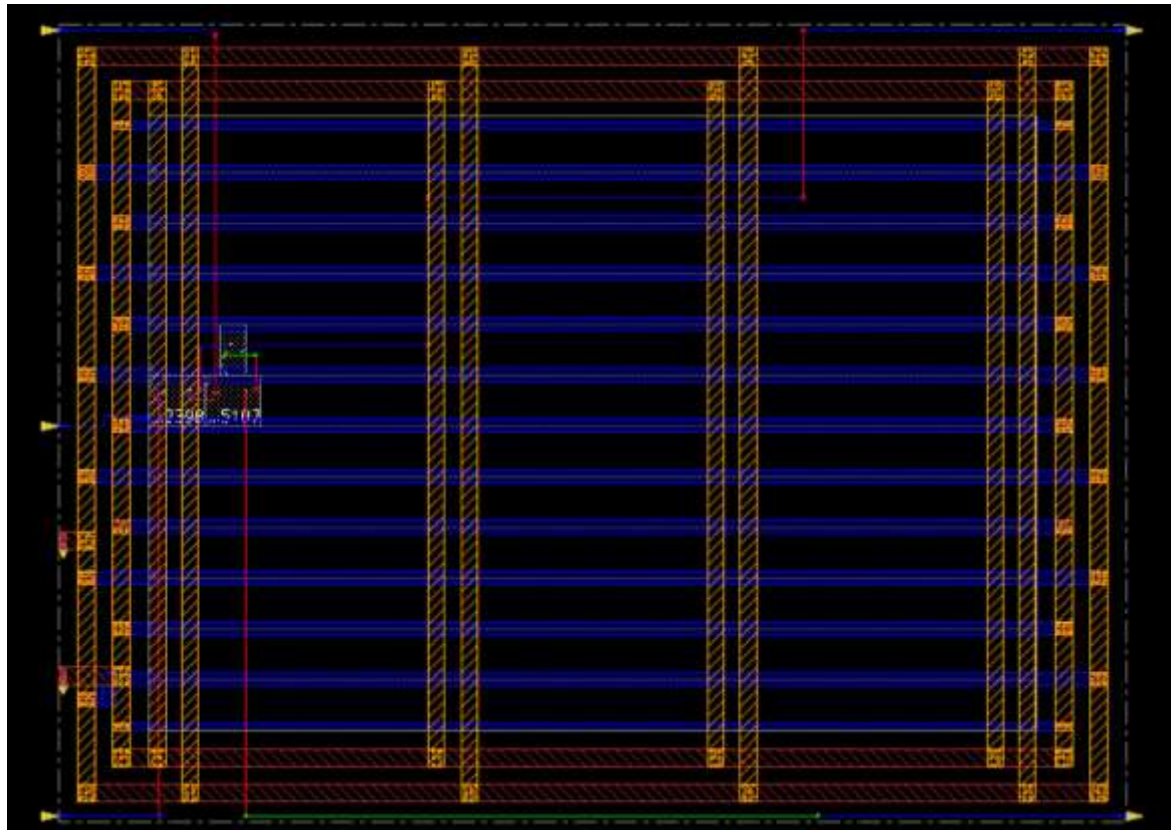


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ROUTING

The main task of routing is to replace all the signal nets in the design with the actual metal layers

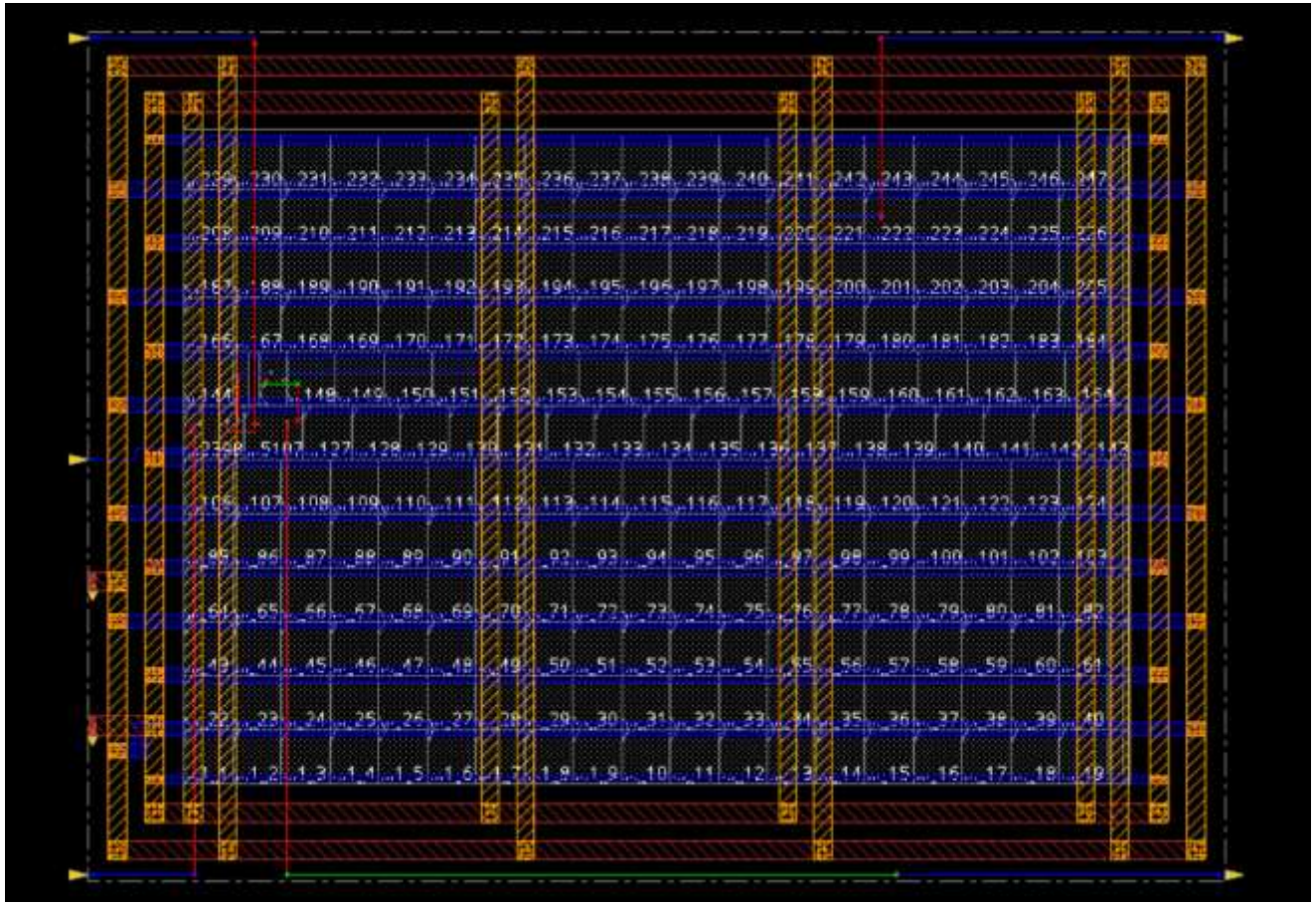
- Click on Route tab
- Now click on NanoRoute



- setAnalysisMode -analysisType onChipVariation

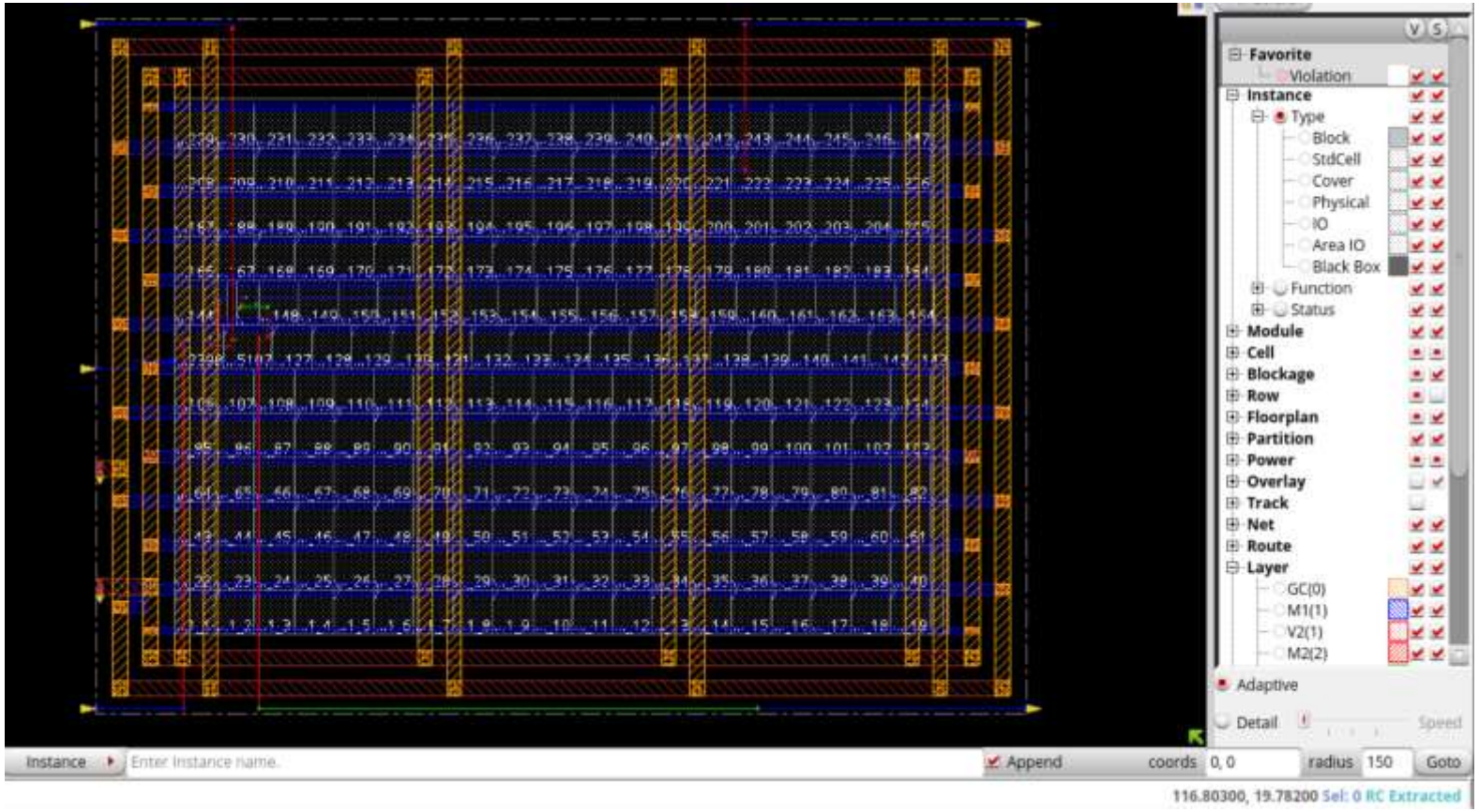
FILLER CELL

- place→physical cell →Add Filler



SIGNOFF VERIFICATION

- RC extraction using Timing -> Extract RC
- Select Verify -> Verify DRC
- Select File -> Save Design
- Select ‘File’ -> ‘Save’ ->’ GDS/OASIS



VERIFICATION

```
*** Starting Verify DRC (MEM: 1814.0) ***

 VERIFY DRC ..... Starting Verification
 VERIFY DRC ..... Initializing
 VERIFY DRC ..... Deleting Existing Violations
 VERIFY DRC ..... Creating Sub-Areas
 VERIFY DRC ..... Using new threading
 VERIFY DRC ..... Sub-Area: {0.000 0.000 118.160 87.920} 1 of 1
 VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.

 Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:00.0  ELAPSED TIME: 0.00  MEM: 3.0M) ***
```

```
innovus 1>
******* START VERIFY ANTENNA ********
Report File: full_adder.antenna.rpt
LEF Macro File: full_adder.antenna.lef
Verification Complete: 0 Violations
******* DONE VERIFY ANTENNA ********
(CPU Time: 0:00:00.0  MEM: 0.000M)
```

STREAM FILE

```
Stream Out Information Processed for GDS version 3:
Units: 1000 DBU

Object                             Count
----------------------------------------
Instances                            252

Ports/Pins                             7
    metal layer M1                     5
    metal layer M4                     2

Nets                                  30
    metal layer M1                    12
    metal layer M2                    16
    metal layer M3                     2

    Via Instances                     22

Special Nets                          59
    metal layer M1                    41
    metal layer M5                     6
    metal layer TOP_M                 12

    Via Instances                    158

Metal Fills                            0

    Via Instances                      0

Metal FillOPCs                         0

    Via Instances                      0

Metal FillDRCs                         0

    Via Instances                      0
```